

MODEL QUESTION PAPER

TED (10) – 5051

Reg. No

(REVISION - 2010)

Signature

FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY

AUDIO AND VIDEO SYSTEMS

[Time:3 hours]

(Maximum marks : 100)

PART ---- A

Marks

I Answer the following questions in one or two sentences. Each questions carries 2 marks.

- 1 What is the principle of magnetic recording?
- 2 What is the function of T V camera?
- 3 What is the advantage of interlaced scanning?
- 4 List the factors affecting tuner design
- 5 What is meant by compatibility of T V transmission? (5x2=10)

PART –B

11 Answer any five of the following. Each question carries 6 marks.

- (1) Explain pressure and intensity of sound waves.
- (2) Explain overtones and timber.
- (3) Draw the block diagram of monochrome T V receiver.
- (4) What are the essential function of IF stage.
- (5) Explain positive and negative modulation
- (6) What is meant by subtractive and additive mixing?
- (7) What is automatic colour control? (6x5=30)

PART – C

(Answer one full question from each unit. Each question carries 15 marks)

UNIT --1

III (a) What is the principle of magnetic recording.

(b) Explain magnetic recording and reproduction of sound. 10

OR

IV (a) With neat diagram explain the construction of moving coil cone type
loudspeaker. 8

(b) Explain the characteristics of moving coil cone type loud speaker. 7

UNIT --11

V (a) Explain positive and negative modulation. 7

(b) Explain the pre equalizing and post equalizing signals. 8

OR

VI(a) Sketch the composite video signal wave form for three successive lines and indicate
(i) extreme white level.(ii) blanking level, (iii) pedestal height,(iv) sync pulse level. 8

(b) Explain vestigial sideband transmission. 7

UNIT -- 111

VII (a) Explain the factors affecting tuner design. 8

(b) Explain tuner operation. 7

OR

VIII (a) Explain the need for R F Amplifier. 7

(b) Draw and explain R F amplifier stage. 8

UNIT - IV

IX (a) Explain the compatibility of T V Transmission. 8

(b) Enumerate essential requirement that must be met to make a a colour system fully
compactable. 7

OR

X (a) Draw and explain Delta gun colour picture tube 8

(b) Explain generations of colour rasters.